

Invariant Planes, Exact Motion, and Null Degeneration of a Constant Electromagnetic Lorentz Flow

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Abstract

We classify proper-time Lorentz motion in every constant electromagnetic field. The characteristic polynomial produces complementary boost and rotation planes, an exact exponential, and an exact position integral. Proper time is never replaced by coordinate time.

1 Convention and invariant factorization

Use $\eta = \text{diag}(1, -1, -1, -1)$ and $\kappa = q/m$. We freeze

$$A = \kappa \begin{pmatrix} 0 & E_x & E_y & E_z \\ E_x & 0 & B_z & -B_y \\ E_y & -B_z & 0 & B_x \\ E_z & B_y & -B_x & 0 \end{pmatrix}. \quad (1)$$

Thus $A^T \eta + \eta A = 0$, and for $u = (u^0, u^0 \mathbf{v})$ the spatial part of $u' = Au$ is $\mathbf{u}' = \kappa u^0 (\mathbf{E} + \mathbf{v} \times \mathbf{B})$. This fixes both electric and magnetic signs. The prime denotes proper time τ , and $x' = u$. Exact constant-field Lorentz evolution has a long lineage; see Chin [1]. We make no priority claim.

Theorem 1 (invariant scales and planes). *There are unique $a, b \geq 0$ such that*

$$\chi_A(z) = (z^2 - a^2)(z^2 + b^2), \quad (2)$$

$$a^2 - b^2 = \kappa^2(|\mathbf{E}|^2 - |\mathbf{B}|^2), \quad a^2 b^2 = \kappa^4(\mathbf{E} \cdot \mathbf{B})^2. \quad (3)$$

If $D = a^2 + b^2 > 0$, then

$$P_h = \frac{A^2 + b^2 I}{D}, \quad P_r = \frac{-A^2 + a^2 I}{D} \quad (4)$$

are complementary projectors, with $A^2 P_h = a^2 P_h$ and $A^2 P_r = -b^2 P_r$.

Proof. A direct determinant of (1) gives $z^4 + \kappa^2(|\mathbf{B}|^2 - |\mathbf{E}|^2)z^2 - \kappa^4(\mathbf{E} \cdot \mathbf{B})^2$; solving its quadratic in z^2 gives (2)–(3). Cayley–Hamilton then verifies every identity in (4). $\square$

2 Exact velocity and position

Write all quotients below by analytic continuation at $a = 0$ or $b = 0$.

Theorem 2 (closed flow). *For $D > 0$,*

$$e^{\tau A} = P_h \left(\cosh(a\tau) I + \frac{\sinh(a\tau)}{a} A \right) + P_r \left(\cos(b\tau) I + \frac{\sin(b\tau)}{b} A \right), \quad (5)$$

$$\begin{aligned} \Phi(\tau) = P_h & \left(\frac{\sinh(a\tau)}{a} I + \frac{\cosh(a\tau) - 1}{a^2} A \right) \\ & + P_r \left(\frac{\sin(b\tau)}{b} I + \frac{1 - \cos(b\tau)}{b^2} A \right). \end{aligned} \quad (6)$$

The solution is $u(\tau) = e^{\tau A} u_0$ and $x(\tau) = x_0 + \Phi(\tau) u_0$. Moreover $\eta(u, u) = \eta(u_0, u_0)$ and $\det e^{\tau A} = 1$.

Proof. Functional calculus on (4) gives (5), and direct integration gives (6). Since $A^T \eta + \eta A = 0$, differentiation of $e^{\tau A^T} \eta e^{\tau A}$ gives zero. Finally $\det e^{\tau A} = e^{\tau \text{tr} A} = 1$. $\square$

References

- [1] S. A. Chin, *Relativistic Motion in a Constant Electromagnetic Field*, arXiv:0809.0859 [math-ph] (2008), [doi:10.1063/1.3064796](https://doi.org/10.1063/1.3064796).